

Introduction to Data Science and AI (Fall 2026, DATA 201)

Schedule
Section 1: MWF 9:00am - 9:50am

Instructor
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ISC4 3313

Office Hours:
Instructor: Friday 11am - 1pm or by appointment.
GTA: TBD (located in the SSC).

Prerequisite(s):
Python, Numpy, and Pandas.

Credit Hours: 3

Course Description

This course provides a practical introduction to data science, covering the full workflow from data preparation to real-world applications. Students will learn how to handle and clean data, explore it through statistical analysis and visualization, and apply core machine learning methods for both supervised and unsupervised tasks. The course emphasizes developing an intuitive understanding of how and why each algorithm works, rather than focusing on heavy mathematical theory, which is addressed in DATA 301. Key topics include model selection, training, testing, evaluation, generalization, regularization, and case studies demonstrating how the data science workflow is used to solve practical problems.

Course Outline

The course outline is provided below. Weekly content may be adjusted as needed to reflect the pace and progress of the class.

Table 1: Course Modules and Approximate Schedule

Module	**Topic**	**Duration**
Module 1	Data Preparation	$\sim$ 2 weeks
Module 2	Learning with Regression Models	$\sim$ 2 weeks
Module 3	Generalization, Regularization, and Selection	$\sim$ 2 weeks
Module 4	Learning with Classification Models	$\sim$ 2 weeks
Module 5	Learning with Tree-Based Models	$\sim$ 2 weeks
Module 6	Learning with Unsupervised Models	$\sim$ 2 weeks
Final Project		

Expectations

All students are strongly encouraged to bring their laptops/notebook computers to class since we plan on having a significant amount of active coding.
Students are expected to attend all lectures regularly and actively engage with the course material. They should ensure that all activities are delivered on time. In addition, students are encouraged to participate in discussions, collaborate respectfully with peers, and take responsibility for keeping up with readings and practice exercises outside of class. Consistent preparation and active involvement will be essential for your success in this course.

Attendance

Attendance is encouraged for this course, as class sessions will include live coding activities, hands-on exercises, and discussions that are designed to support student learning and project development. Regular participation will help students stay engaged with the course material and successfully complete course assessments.

Classroom Behavior

Students are expected to engage respectfully during discussions to support the open exchange of ideas. All perspectives are welcome, and students should practice active, attentive listening to maintain a supportive learning environment. Disruptive behaviors, including loud conversations with peers or other actions that interfere with class activities, are not permitted. Eating during class sessions is not allowed in order to help maintain a clean, focused, and respectful classroom environment.

Blackboard Course Management System

Blackboard will be used as the primary learning management system for this course. Course materials, including lecture slides, challenges, quizzes, and grades will be posted on Blackboard.

Communication Platforms

Course communication will take place through the following platforms:

- **Blackboard Announcements:** Important updates, reminders, and schedule changes will be posted through Blackboard announcements.
- **Module Agendas:** Each module includes an agenda outlining the planned lectures, activities, assessments, and deadlines. Students are responsible for checking module agendas regularly, as they may be updated due to academic, environmental, technological, or other circumstances.
- **Module Discussions:** Each module includes a Questions and Discussion forum where students may ask questions about course content, quizzes, challenges, and projects. Students should review existing posts before submitting a new question.
- **Email:** Email should be used for individual or confidential matters, including grades, absences, accommodations, extensions, or other personal concerns.
- **Office Hours:** Students may attend office hours for questions requiring detailed explanations or individualized assistance.

Students are responsible for regularly checking Blackboard, course announcements, and module agendas. They should not post assessment answers, challenge solutions, personal information, or grade-related questions in module discussions. Only information provided by the instructor or GTA should be considered official course guidance. Please allow up to two business days for a response.

Communication Policy

For all questions or concerns related to quizzes or challenges, students should first communicate with the Graduate Teaching Assistant (GTA). The GTA is the primary point of contact for course-related issues and is equipped to address most questions. If an issue cannot be resolved at that level, the GTA will escalate the matter to the instructor as appropriate. Adhering to this communication process helps ensure fairness, consistency, and efficient course administration. Confidential or sensitive matters may be communicated directly to the instructor.

When sending emails to the course staff, students must include the course number (DATA 201) and their section number (Section 1) in the **email subject line**. This helps ensure that messages are properly routed and addressed in a timely manner.

Please give 48 hours (usually 24 hours or less) for faculty to respond to email on weekdays. Emails sent on the weekend will be responded to on the following Monday.

Student Accessibility Services (SAS)

William & Mary accommodates students with disabilities in accordance with federal laws and university policy. Any student needing accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services at 757.221.2512 or at sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation.

Student Success Center (SSC)

Beginning next week, the Student Success Center (SSC) will be available to all Data Science students on weekdays from 10:00 a.m. to 8:00 p.m. It is located on the third floor of ISC 4. After exiting the elevator, the center is immediately on the right.
Graduate and undergraduate tutors are available to:

- Assist with any Data Science course
- Answer general academic and orientation questions
- Support students in navigating the Data Science program

The SSC also provides a well-equipped and supportive environment for independent or collaborative study.
All students should check in before using the center, whether they are seeking assistance or using the space for self-study. To check in, scan the SSC QR code and complete the brief form, which takes approximately 30 seconds.

Mental & physical well-being

There are many resources to help navigate emotional, psychological, physical, medical, and material concerns, including:

- Counseling Center at (757) 221-3620. Services are free and confidential.
- Health Center at (757) 221-4386.
- The free Timely Care app gives students access to 24-7 remote counseling support.

Assessment Components

Quizzes

At the end of each module, students will complete a quiz designed to assess their understanding of the key concepts covered in that module. There will be a total of **six quizzes**. Each quiz will consist of multiple-choice questions and an essay question.

Challenges

Throughout the course, students will complete **four challenges** designed to apply the concepts learned in class to practical and creative problems. These challenges will involve using computational methods to develop data analyses, visualizations, predictive models, or analytical workflows.

- **Pair work:** Each challenge will be completed in pairs. Both students are expected to contribute meaningfully to the work and understand the entire submission, regardless of how the tasks were divided.
- **Use of GenAI:** Students are expected to use generative AI tools to support the development of **advanced** workflows and applications. The use of GenAI must be disclosed, including the tools used and a brief explanation of how they contributed to the work. Students remain responsible for verifying the accuracy and quality of all AI-generated content.
- **Submission:** Each pair will submit one shared solution. The submission will be evaluated based on implementation, problem-solving, interpretation of results, critical evaluation of the solution, and overall quality.
- **Oral understanding check:** After submitting each challenge, both students must attend a brief, scheduled meeting with the TA during designated SSC tutoring hours. Each meeting will last approximately **7–10 minutes**.
- **Individual questions:** Each student will answer at least one question about the submitted work. Students may be asked to explain any part of the submission, including the methodology, code, results, design decisions, GenAI contributions, limitations, or possible improvements.
- **Grading:** The submitted work will receive a shared grade, while each student will receive an individual grade for the oral understanding check. The oral check will account for **20%** of each challenge grade.
- **Individual accountability:** Because individual oral grades may differ, both partners should review the complete submission and be prepared to explain and justify the work independently. A student who cannot adequately explain the submitted work may receive a reduced individual grade and may be asked to complete an additional follow-up assessment.

Final Project
At the end of the course, students will complete a final project that integrates and applies the knowledge and skills developed throughout the semester. Students will work in pairs to select a meaningful and appropriate data science problem and address it through data preparation, model development, evaluation, and interpretation.
The final project will include the following components:

- **Project selection:** Each pair will select and formulate a meaningful, appropriate, and sufficiently challenging data science problem.
- **Written report:** Each pair will submit a report describing the problem, data, methodology, implementation, results, limitations, and conclusions.
- **Recorded presentation:** Each pair will submit a short recorded presentation summarizing the project, explaining the main decisions, and presenting the most important results.
- **Use of GenAI:** Students are expected to use generative AI tools to support the development of advanced workflows and applications. The use of GenAI must be disclosed, including the tools used and a brief explanation of how they contributed to the project. Students remain responsible for verifying the accuracy and quality of all AI-generated content.
- **Online oral defense:** After submitting the project, both students will attend a brief online meeting and answer questions about their work. Each student will be asked at least one question and may be asked to explain any part of the project, including the problem formulation, methodology, code, results, design decisions, GenAI contributions, limitations, or possible improvements.
- **Individual accountability:** Both partners must contribute meaningfully and understand the entire project, regardless of how the work was divided. The project deliverables will receive a shared grade, while performance during the oral defense will be evaluated individually.
- **Evaluation:** Students will be evaluated on the choice and formulation of the problem, methodology, implementation, analysis and interpretation of results, critical evaluation of the solution, quality of the report and presentation, and their ability to explain and defend the work.

Grade Distribution

>= 93.00	A	73.00 - 76.99	C
90.00 - 92.99	A-	70.00 - 72.99	C-
87.00 - 89.99	B+	67.00 - 69.99	D+
83.00 - 86.99	B	63.00 - 66.99	D
80.00 - 82.99	B-	60.00 - 62.99	D-
77.00 - 79.99	C+	<= 59.99	F

Quizzes	30%
Challenges	35%
Final Project	35%

Quiz Policy

Quizzes will be administered during scheduled class sessions unless otherwise stated. Students who miss a quiz must complete it during the next available office hours and will receive a **10% penalty**. Failure to do so will result in a grade of zero. Exceptions will be considered only in documented extenuating circumstances.

Challenge/Project Submission Policy

Challenges and the final project are expected to be submitted by the stated deadlines. Late challenges and projects will incur a **10% penalty per day for up to three days**. Submissions more than three days late will not be accepted and will receive a grade of zero. Exceptions will be considered only in documented extenuating circumstances.

Academic Integrity

All class-related activities are governed by the William & Mary Honor Code.

Use of GenAI

The use of generative AI (GenAI) is not permitted during quizzes unless explicitly stated otherwise by the instructor.
Students are expected to use GenAI tools for challenges and the final project to support the development of more **advanced** workflows, analyses, and applications. However, students remain fully responsible for the accuracy, quality, and originality of their submitted work. They must understand the methods and code used, critically evaluate AI-generated content, and be able to explain and justify their decisions, implementation, and results.
Students may be asked to discuss or demonstrate their work and explain the process used to produce it. Submitting work that a student does not understand or cannot explain, or using GenAI where it is prohibited, may constitute an Honor Code violation and may be submitted for disciplinary review.

Important Dates

- Add/drop deadline: September 4.
- Labor Day (no classes): September 7.
- Fall Break: October 8–11.
- Withdrawal deadline: October 26.
- Election day (no classes): November 3.
- Remote Instruction Days (Courses must be delivered remotely, offices closed): November 23–24.
- Thanksgiving (no class): November 25-29.
- Last day of classes: December 4.
- Final Project deadline: TBD.

Recommended Resources

Hands-On Machine Learning with Scikit-Learn and PyTorch, by Aurélien Géron. O'Reilly.